

The research group “Sensory Dynamics and Behaviour” of Prof. Dr. Tobias Ackels at the Institute of Experimental Epilepsy and Cognition Research at the University Hospital Bonn invites applications for a position as:

Postdoctoral Researcher (m/f/d)

The position is initially limited to three years (TV-L E13/100%),
starting as soon as possible.

Project overview:

The successful candidate will join an interdisciplinary research program investigating how the olfactory system performs **stimulus selection and figure-background segregation under naturalistic, temporally complex conditions**. Natural odour plumes are turbulent and intermittent; their rapid temporal structure conveys spatial and contextual information about odour sources but strongly challenges the extraction of behaviourally relevant foreground odours from fluctuating backgrounds.

The project will dissect how olfactory bulb and cortical circuits encode and select foreground odour “objects” from complex backgrounds by combining high-speed, plume-like odour stimulation with *in vivo* large-scale recordings (Neuropixels in head-fixed and freely moving mice, 2P/miniscope imaging), optogenetic and chemogenetic manipulation of cortical feedback, and quantitative behavioural assays.

This work addresses a fundamental computational question in mammalian olfaction, links to broader topics such as sensory selection, active sensing, and predictive coding, and is embedded in ongoing experimental and computational collaborations in Bonn.

Responsibilities:

- Design, conduct, and analyse *in vivo* physiology experiments.
- Perform Neuropixels recordings in head-fixed and freely moving mice, and/or 2P/miniscope imaging to probe single-cell and population dynamics during dynamic odour processing.
- Conduct quantitative analysis of large-scale neural population data (decoding, dimensionality reduction, temporal analysis, etc).
- Use optogenetic and chemogenetic tools to manipulate defined feedback pathways and test their causal role in dynamic stimulus selection.
- Prepare manuscripts, present at international conferences, contribute to collaborative meetings, and help train junior researchers and maintain custom-built experimental setups.

Requirements:

- PhD in neuroscience, biology, physics, engineering, or related.
- Experience with rodent behaviour and/or neural recordings (extracellular electrophysiology calcium imaging and/or miniscopes).
- Proficiency in at least one scientific programming environment (Python, Matlab) and enthusiasm for quantitative data analysis.
- Interest in setting up, working with, and understanding custom-built equipment.
- Excellent written and spoken English and a professional, team-oriented, and flexible working attitude driven by curiosity and ambition.

Experience with large-scale electrophysiology (e.g. Neuropixels), head-fixed or freely moving behaviour, or olfactory research is highly advantageous but not strictly required.

We offer:

- A cutting-edge research environment focused on sensory coding, olfactory circuits, and active perception, with strong links to computational neuroscience.
- Access to state-of-the-art physiology platforms including 2P imaging, Inscopix miniscope, optogenetics, closed-loop behaviour, and Neuropixels recordings.
- Close supervision and scientific mentoring by the PI, regular group meetings, and support in developing independent research directions and competitive fellowship applications.
- Opportunities to participate in international conferences and workshops.
- A vibrant and highly collaborative international research community in Bonn, with excellent local infrastructure and quality of life.
- Employment according to TV-L (EG13, 100%) with supplementary benefits (VBL pension plan), corporate customer ticket (subsidized public transport), and access to on-campus daycare facilities.
- We welcome applicants from all backgrounds and identities.

How to apply:

Applicants should send their application in **a single PDF file** (max. 5 MB), including

- Cover letter outlining research interests and motivation
- CV
- Contact information for two academic references

Applications will be reviewed on a rolling basis. Informal inquiries are welcome. Please send your application to Prof. Tobias Ackels via email (tobias.ackels@ukbonn.de).

Application deadline: **28.02.2026**

For further information, please visit our lab website or contact Prof. Ackels directly.

Prof. Dr. Tobias Ackels

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